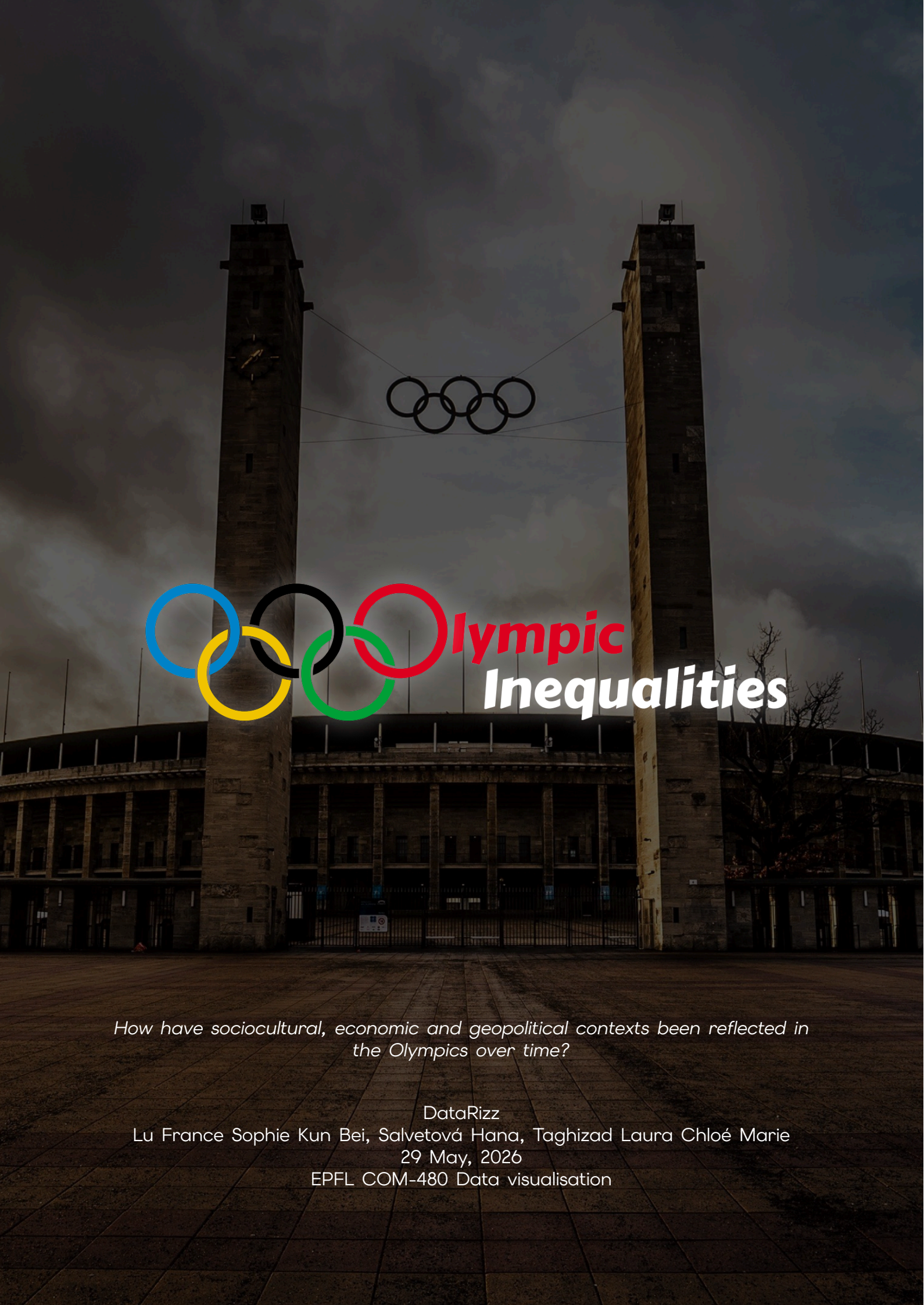


Olympic Inequalities

How have sociocultural, economic and geopolitical contexts been reflected in the Olympics over time?

DataRizz

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EPFL COM-480 Data visualisation

OUR PATH

When choosing our dataset, we wanted to work on a hot topic. In February, the Milano Cortina 2026 took place. Although 47% of the athletes were women, only men could compete in the Nordic Combined discipline. In addition, given the current geopolitical conflicts, some countries were not allowed to join the Games. We therefore decided to address the following main research question:

How have sociocultural, economic and geopolitical contexts been reflected in the Olympics over time?

We found datasets on Kaggle up to 2016, merged and cleaned them into a single file. We then undertook exploratory data analysis. We analysed the following topics:

- Summer / winter participating countries over time;
- Summer / winter gender representation over time;
- Summer / winter sports participation over time;
- Summer / winter top countries by medal count;
- Economic inequality (GDP per capita vs. medal count over time);
- Athlete characteristics by age and medal type;
- Countries in conflict during Olympic years.

At this stage, we decided that our visualisations should demonstrate Olympic inequalities across five axes: delegations, gender, athletes, geopolitics and the GDP and population of countries. We used Excalidraw to brainstorm sub-research questions and sketch some initial visualisations for our website skeleton.

Firstly, we wanted to use a Sankey diagram to show which **delegations** / continents won medals in which disciplines over time. Secondly, we wanted to visualise how **gender** participation differs across sports and disciplines. Both visualisations are split between Summer and Winter Olympics. Thirdly, we wanted to show the body types of medalled **athletes** based on their weight and height, using MakeHuman. Fourthly, we wanted to visualise the **geopolitical** conflicts on a map and analyse their influence on the participation of the relevant delegations in the Olympics using a bar plot. Last but not least, we wanted to compare the fair medal count of two countries. We define fairness by rescaling the number of medals won based on the average **GDP** and **population** of the delegations.

When designing the skeleton of our website, we looked at projects from previous years and conducted online research. We chose fullPage.js, a JavaScript library, to build full-screen one-page scrolling websites with horizontal slides. We created a new folder in our repository to contain it. We added the sketches and chose initial fonts for the headers and body texts.

Next, we used D3.js along with AI tools to develop the initial visualisations and integrate them into the website, replacing the sketches. We then improved the athlete models in the body type section. At the same time, we made the visualisations interactive and clickable for a better user experience. We also added some features and new visualisations, which will be discussed in detail in the next section. We then worked on standardising the styling and fixing minor visual discrepancies. In parallel, we updated the sketches to incorporate our new ideas. We further worked on integrating more storytelling text. Finally, we prepared a screencast to present our website and this process book about it. We used Figma to design the stickman race in this document.

VISUALISATIONS & CHALLENGES

Following the structure of our website, we now discuss the different visualisations, the design decisions we made, and the challenges we faced. Firstly, we decided it would be more logical to put the gender section before the medalled countries. We changed the body font and styled the contextual text of the topics in columns so that lines are neither too wide nor too short.

We had trouble fighting back unnecessary features implemented by AIs, which we solved by manually deleting and restyling the code snippets. Initially, we used a single 42.5 MB data file for all plots, which made our website very laggy. Therefore, using the file `extract_viz_data.ipynb`, we created individual extracted data files to be loaded in each visualisation, which greatly improved the latency. After that, we also managed the displayed loading texts by slowing down the connection while testing.

Running Stickman & Track

To help users keep track of the different visualisations and **themes**, we had a new idea of using a **stickman** for storytelling. 🏃 runs along a vertical track going through a warm-up area, starting blocks, refreshment stands, a finish line, and an area to cool down. This allows 🏃 to slowly discover the dataset. While implementing 🏃, we had a few issues regarding its smoothness and correct positioning while staying synchronised with the different slides. The warm-up and cool-down SVGs were added in a second stage and were taken from the [Noun Project](#).



Warm Up

🏃 first warms up on a designed 'home page' to get familiar with the theme. After briefly brainstorming on the design of the **logo**, we used Figma to develop it. We wanted to integrate the title 'Olympic Inequalities' as well as the Olympic Rings. We also designed the favicon in Figma using a Noun Project icon recolored to match the site's gold theme and exported as a .ico file. Compared to our initial website skeleton, we only kept the background image behind the logo and opted for a black background elsewhere.



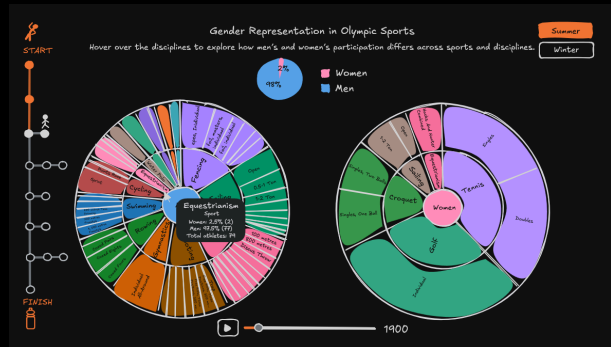
Starting Blocks

🏃 gets to the **starting blocks** and prepares for the run. This section outlines the main content and data of the website. The different stops correspond to the various visualisation topics: gender, medals, bodies, geopolitics and fairness.

Gender

🏃 starts running and stops at the first refreshment stand. The drink that the stickman gets introduces the context of this topic: the intertwining of gender and the Olympics. Swiping right, we created a season selector button and a play / pause button on that slide to see the evolution of gender and sports over time. Using D3.js, we implemented a pie chart to show the proportion of men vs. women participating each year. Being old-school, we used the stereotypical gender colours. Below that, we integrated two sunburst plots, one for men and one for women, to reveal the participation in different sports and disciplines. For the same sports, a lighter colour represents a lower percentage of participation of that gender.

When hovering over a sport or discipline, we added a tooltip with the detailed numbers and percentage of men vs. women participating. We synchronised the year for the 'Summer' and 'Winter' Games to make them comparable when switching between them. However, we struggled to size the sunbursts such that they would not overlap but still be big enough to be readable.

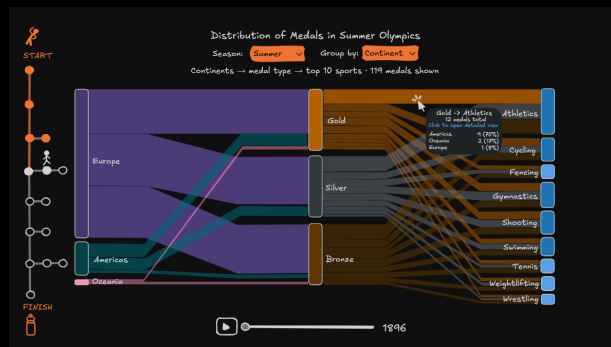


Medals

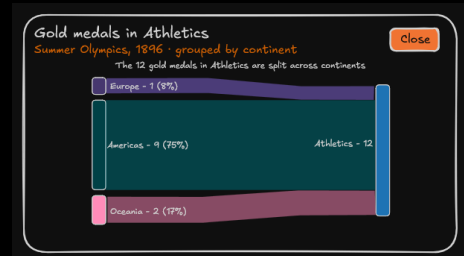
🏃 runs to the next refreshment stops, which introduces the distribution of **medals** among the delegations. We were originally curious about which countries tend to win more medals regularly. After thought, we added a new visualisation showing non-medal-winning delegations on a map with their statistics.

Medalled

Swiping right, we set selectors for the season and grouping mode on that slide. We added a play / pause button to see the evolution of the **distribution of medals across delegations** over time. We kept the original Sankey diagram, but switched the medal to the centre and the sports to the right, to be more logical. We only display the top 10 sports with the most disciplines.

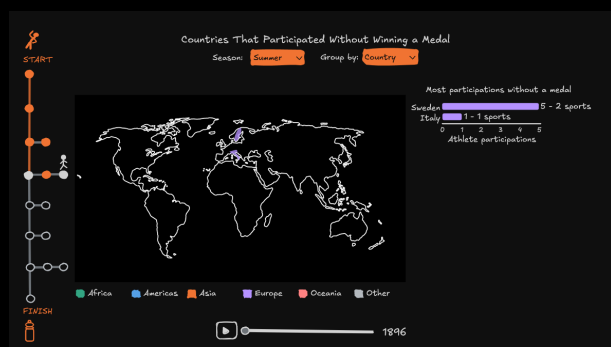


Hovering on a flux on the right side of the plot, it shows a tooltip preview of the detailed number and percentage of the distribution. Clicking on it opens a pop up window with the Sankey view of that link. We synchronised the selectors of this visualisation with the non-medalled one.



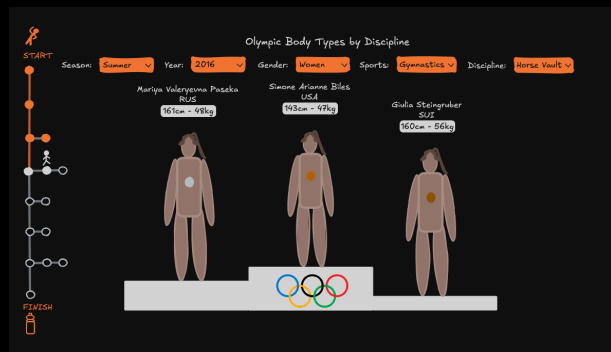
Non-Medalled

Swiping right a second time, we added this new visualisation. We have the same selectors and play / pause button to see the evolution of **countries that participated without winning a medal**. The map highlights these countries in colour. Below that, we added a legend showing the continents, which is useful when grouping by country. The horizontal bar plot shows the top 10 non-medalled delegations in terms of highest athlete participation. We added the exact number of participants beside them. We had some difficulties sizing the map and making the countries visible enough against the black background.



Body Types

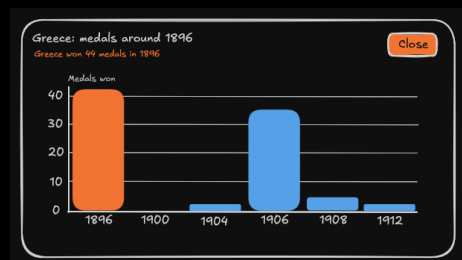
🏃 runs to the next refreshment stand, where we discuss **body types** per disciplines in the Olympic. Swiping right, we created, on that slide, selector fields for season, year, gender, sports and discipline. Below that, we display the podium with the three 'physiques' of the medalled athletes, scaled by height and weight. Their name, delegation, height and weight are displayed above.



Initially, we used D3.js non-realistic body and medal models. We then used MakeHuman to have better-looking avatars and Blender to integrate the medals. We saved one GLB model per gender and per medal, which were challenging to integrate into the website. Positioning the camera and scaling the models were also difficult to implement. We managed to set it so that there is not too much spacing below the podium. We also had issues regarding the athletes' names and details cropping the heads of, e.g., men basketball athletes. We also had trouble with the podium, which we build by stacking blocks on top of each other. The top colour of the bottom block would appear on the front side, making an unwanted line appear.

Geopolitical Conflicts

🏃 runs to the next refreshment stand to explore the influence of **geopolitical conflicts** on the Olympic Games. Swiping right, we highlight countries on the map related to a selection of geopolitical and socioeconomic events. On the right is a clickable list of such disruptions. Hovering over a concerned country, we display a toolbox explaining its relation to the event. Clicking on it opens a pop up window showing the evolution of the medals won by that delegation around the year of that geopolitical disruption.



Fair Distribution

🏃 runs to the last refreshment stops, where the **fair** medals computation is based on the average population and GDP. We were originally curious to compare how two countries would perform with fair resources. Later, we thought it would be interesting to show the actual vs. fair rankings of all countries. We use the same extracted data for both of the below visualisations.

Fair Comparison of Countries

Swiping right, we set on that slide selectors for the season, the year and the **two countries to be fairly compared**. We then display for each of them their actual gold, silver and bronze medal counts. We originally had bar plots for that, for which we had troubles setting the max value to be constant for each subplot. Later, we had the great idea of displaying ‘actual’ medals, which are in reality just small medal-like round dots. We saw that Liechtenstein would have close to 2000 fair gold and silver medals in the Winter Olympics of 1980. The display was impossible with 25 medals per line. Hence, we fixed it by displaying the first 50 medals and then showing the rest of the medals count underneath them.



At the bottom of the slide, we added the specific formula we used to compute the fair values. In another extracted data, we precomputed the fair gold, silver and bronze medal counts normalised to the mean population and mean GDP. If a country won no medal, we took the fair count of the delegation with the closest GDP and population.

Fair Ranking of Countries

Swiping right, we have this new visualisation, with the selectors season and year, to show **actual and fair ranking of countries**. By hovering the line of a delegation, we display a toolbox containing its population and GDP. When clicking, we made it highlight the country in both rankings and scroll to have it visible on both plots. It can be deselected. When scrolling on the tables to have a look at other countries, the website would skip to the ‘Finish Line’ section. We fixed this issue by making these parts non-interactive with sections’ scroll.



Finish Line

🏃 has reached the finish line and finished the run. Wouhou! We **conclude** on our main topics around the Olympics and their visualisations that the Games mirror global inequalities.

Cool Down

🧊 can cool down from the run and drink some water by going to the **footer**. We named it as is to end the story of the stickman’s run by letting the user know who is behind the design of the Olympic Inequalities website.

PEER ASSESSMENT

Lu France Sophie Kun Bei

- Website base using fullPage.js
- Logo and icon design
- Debug visualisations and styling
- Shorten lag of website
- Final sketches on Excalidraw
- Process book & story telling
- Screencast recording

Salvetová Hana

- All base visualisations using D3.js
- Features on visualisations
- Improve visualisations and interactions
- Running stickman base
- Website text
- Repository README.md
- Screencast audio

Taghizad Laura Chloé Marie

- MakeHumans and Blender for body models
- Integrate body models into website
- Improve stickman smoothness & style
- Debug visualisations and styling
- Load messages
- Website story telling
- Screencast video & audio editing